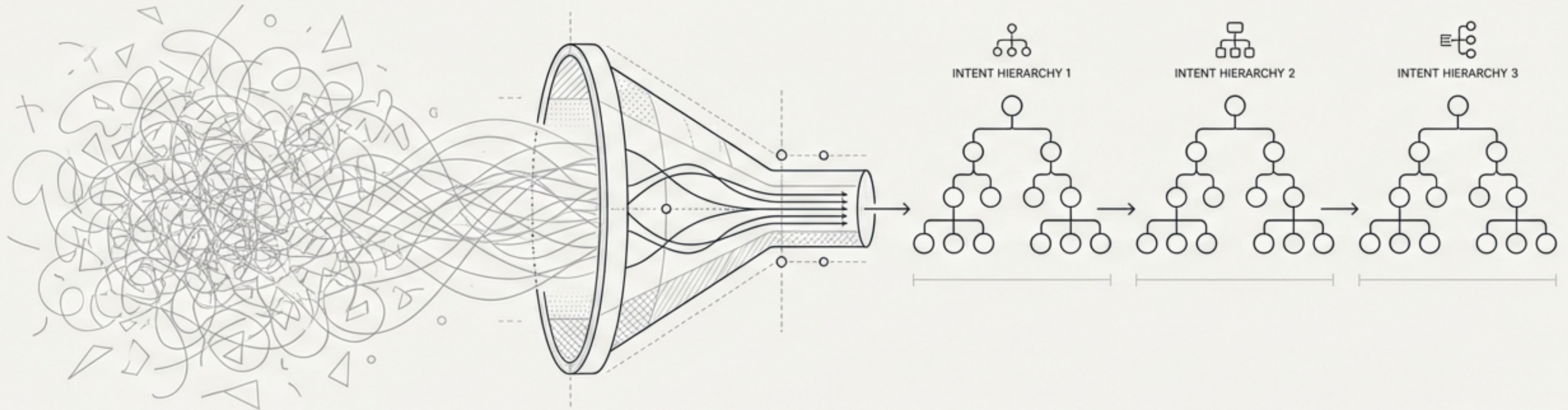


Intelligent Intent Expansion: The V2 Pipeline

An Evidence-Based System for Managing
Conversational AI Complexity



The Core Challenge: Balancing Specificity with Simplicity

Conversational AI platforms often use broad intents, missing specific user needs. The critical challenge is identifying these gaps without over-fragmenting the intent space, which can confuse classifiers and increase costs.

- “Each word being passed to the model should be weighed in gold. If we create ‘ingredients’, ‘components’, ‘material’ separately, we create the possibility for LLM failure.”

Specificity

- ✓ Leads to better user experiences.
- ✓ Enables targeted, accurate responses.

Simplicity

- ✓ Crucial for smaller, efficient model classifiers.
- ✓ Maintains clear, non-overlapping intent boundaries.

Our Goal: A system that achieves specificity without sacrificing simplicity.

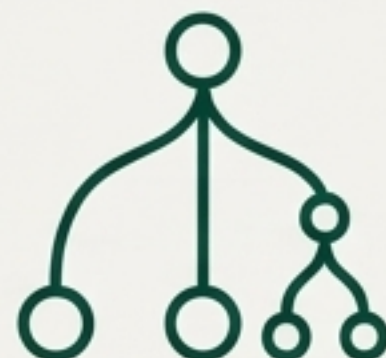
Our Approach is Guided by Three Core Principles



1. AI-Driven Discovery

Instead of relying on hardcoded rules, we leverage LLMs to discover emergent user themes directly from raw messages.

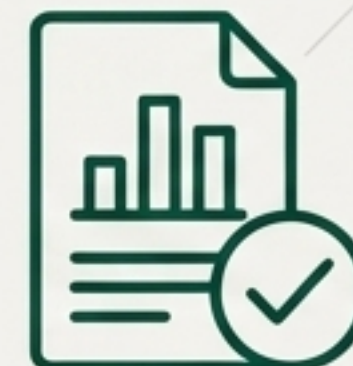
Benefit: Scales effortlessly across new domains (e.g., beauty, fragrance, automotive) without recoding.



2. Hierarchical Intelligence

Our most critical innovation. The system intelligently determines if a new theme should be a new primary intent, a secondary intent under an existing primary, or a refinement.

Benefit: Controls intent growth and prevents prompt bloat.



3. Evidence-Based Recommendations

Every suggestion is a data-backed proposal, not a guess. We provide quantitative and qualitative proof for every decision.

Benefit: Enables confident, low-risk implementation decisions based on a clear verdict (e.g., `STRONGLY_RECOMMEND`).

The Key Innovation: Controlled Growth Through Hierarchical Intelligence

The pipeline's primary function is not just to find new intents, but to place them **correctly within the existing structure**. This prevents uncontrolled fragmentation.

✗ Uncontrolled, Flat Growth

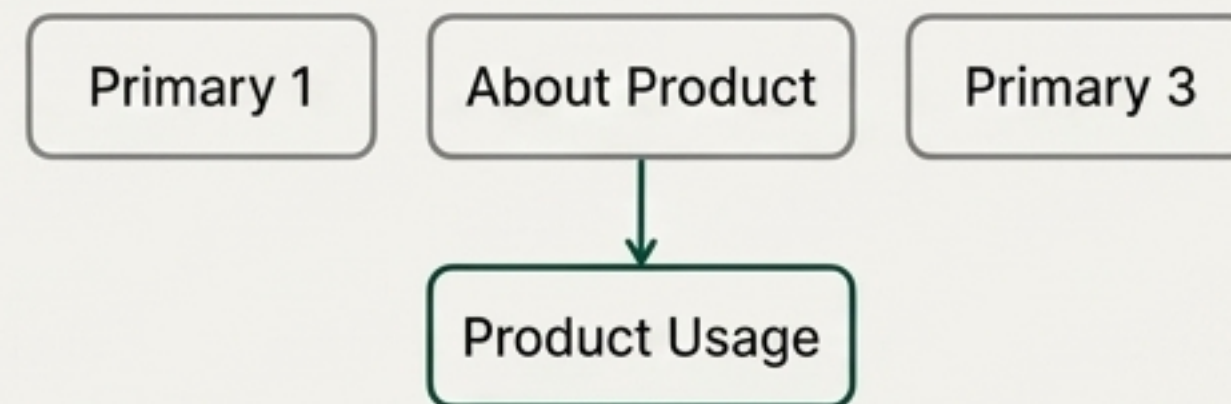


Treating 'Product Usage' as a new **PRIMARY** intent.

****Result****: 3 primaries → 7 primaries.

****Impact****: Prompt **size doubles**, increasing cost and model confusion.

✓ Controlled, Hierarchical Growth

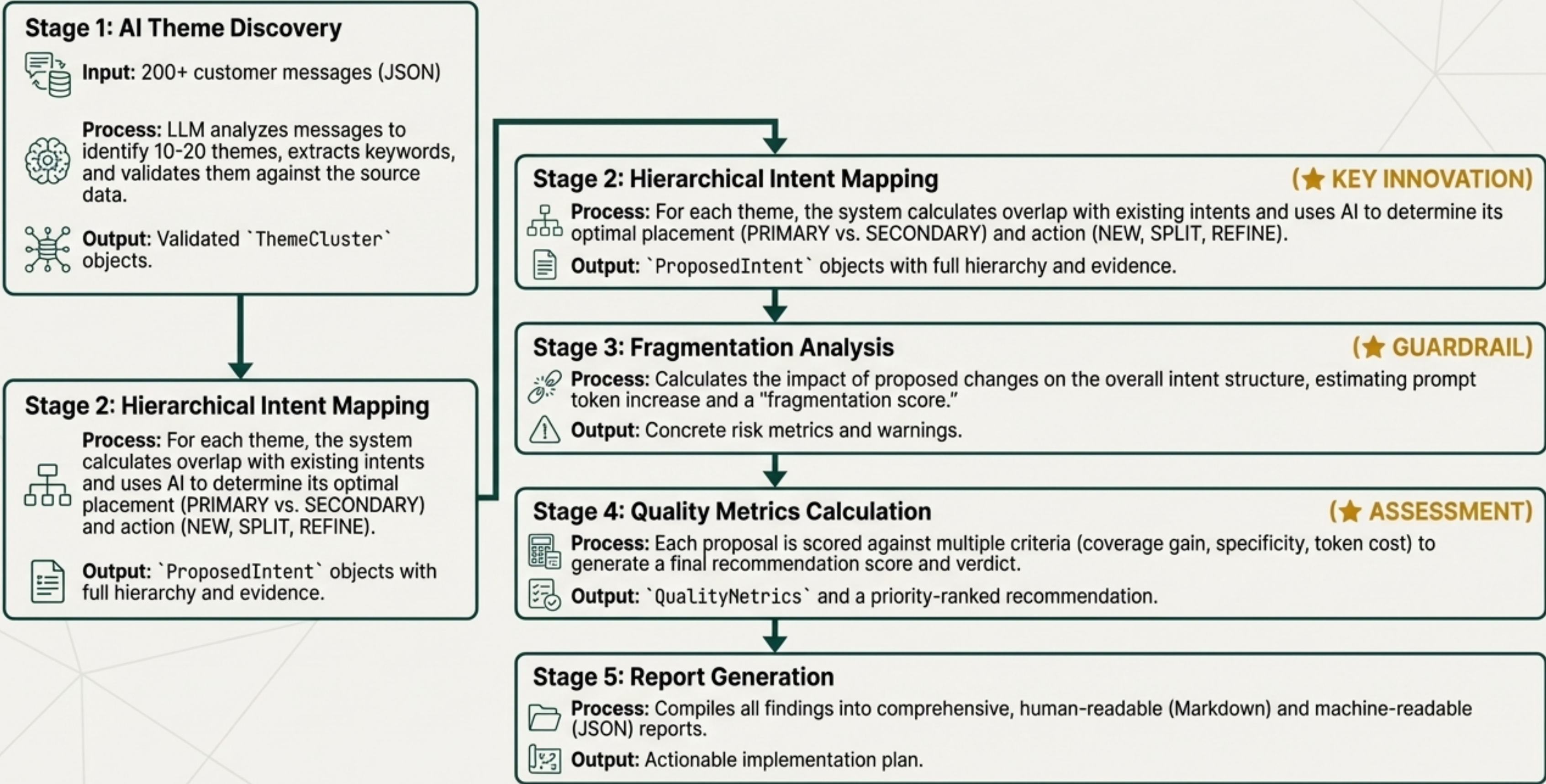


Placing 'Product Usage' as a **SECONDARY** intent under 'About Product'.

****Result****: 3 primaries → still 3 primaries.

****Impact****: Intent structure grows logically without increasing primary-level complexity.

The V2 Pipeline: A 5-Stage Architecture for Intent Discovery and Governance



Every Proposal is Evaluated by a Multi-Criteria Scoring Engine

Single metrics can be misleading. Our system **balances multiple factors** to produce a holistic and reliable recommendation score (0-100).



Findings: High-Priority Intents Discovered in the Beauty Dataset

Analysis of 200 real customer messages in the Beauty/Skincare domain.

Proposal 1: Order Tracking

Verdict	STRONGLY RECOMMEND (Score: 85/100) ✓
Placement	NEW SECONDARY intent under `order_management`.
Evidence	21 messages (10.5% of total). Sample queries: "How to track order?", "Where is my order?".
Justification	The most common post-purchase query. Current `Order Management` intent is too broad and requires a dedicated flow.
Impact	Low fragmentation risk, +10.5% coverage gain, +2.8% token cost.

Proposal 2: Product Recommendation Requests

Verdict	STRONGLY RECOMMEND (Score: 85/100) ✓
Placement	NEW SECONDARY intent under `specific_product`.
Evidence	24 messages (12.0% of total). Sample queries: "Best for oily skin", "Which facewash is good for brighter face".
Justification	Second most common query. Different user need than general info; enables a personalized recommendation engine.
Impact	Low fragmentation risk, +12% coverage gain, +3.0% token cost.

The Pipeline Adapts: Uncovering Domain-Specific Needs in Fragrance

Analysis of 191 messages from the Fragrance/Perfume domain. The system successfully identified both universal e-commerce patterns and unique industry-specific language.

Proposal 1: Product Characteristics - Scent, Longevity, Sillage

Verdict: **STRONGLY RECOMMEND (Score: 80/100)** ✓

Placement: NEW SECONDARY intent under `specific\_product`.

Evidence: 19 messages (9.95% of total). Sample queries: 'How is the sillage?', 'Longevity kitna hai?'.

Domain Insight: The pipeline learned fragrance-specific vocabulary ('sillage,' 'longevity') that is critical for purchasing decisions in this vertical.

Proposal 2: Product Authenticity & Comparison Concerns

Verdict: **RECOMMEND (Score: 55/100)**

Placement: NEW SECONDARY intent under `specific\_product`.

Evidence: 6 messages (3.14% of total). Sample queries: 'Original hai ya fake?', 'Attar vs Perfume spray'.

Domain Insight: Identified a key customer concern in the fragrance market (authenticity), which requires a specific trust-building response.

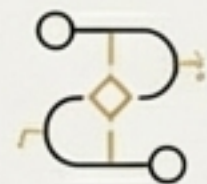
Designing for Reality: How the Pipeline Handles Failure Cases

A robust system anticipates and manages real-world challenges. Here's how V2 handles common failure modes.

1. Failure: API Rate Limits



What Happens: The LLM API call for parent mapping fails (e.g., `429 You exceeded your current quota`).



Pipeline Response: The system detects the failure, logs a warning, and automatically **falls back to a heuristic placement algorithm** based on keyword overlap. The pipeline continues without crashing.

2. Failure: Insufficient Data (<50 Messages)

What Happens: Small datasets produce statistically insignificant patterns.

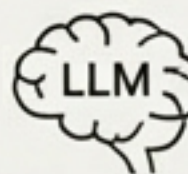


Pipeline Response: The system **issues a warning** in the report if <100 messages are analyzed. It automatically **filters out themes with fewer than 3 messages** and gives low-confidence proposals a **NOT_RECOMMENDED** verdict.

3. Failure: Multilingual Keyword Matching



What Happens: AI discovers Hindi keywords (e.g., 'track करना है'), but fuzzy matching struggles with script variations (Roman vs. Devanagari).

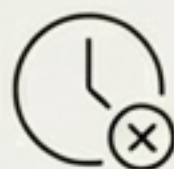


Pipeline Response: The system **still captures ~70% of bilingual messages** through case-insensitive matching. The limitation (potential undercounting) is acknowledged, with a future enhancement planned for LLM-based semantic matching.

Engineering with Foresight: Acknowledged System Limitations

The V2 pipeline is a powerful batch analysis tool. For full production integration, these factors must be considered.

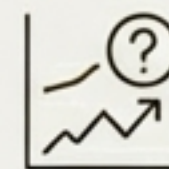
No Real-Time Learning



The pipeline is batch-only and does not learn from live classification feedback or detect intent drift in real time.

Workaround: Run the pipeline on a recurring schedule (e.g., monthly) to analyze new message batches.

No Pre-implementation Accuracy Prediction



The system recommends intents based on evidence volume, not a simulated accuracy score (e.g., confusion matrix).

Workaround: Manual A/B testing is recommended after implementation to validate performance gains.

Token Costs at Scale



Analyzing 1M+ messages can become costly (\$200-\$1000/month).

Mitigation: The system supports batch processing with sampling (e.g., analyze 10% of messages) and can use cheaper models for discovery.

Requires Human-in-the-Loop



The pipeline cannot understand business constraints or implementation costs.

Requirement: A product manager must review high-priority proposals to provide business context and make the final implementation decision.

The Governance Engine: Proactive Guardrails Prevent Bad Suggestions

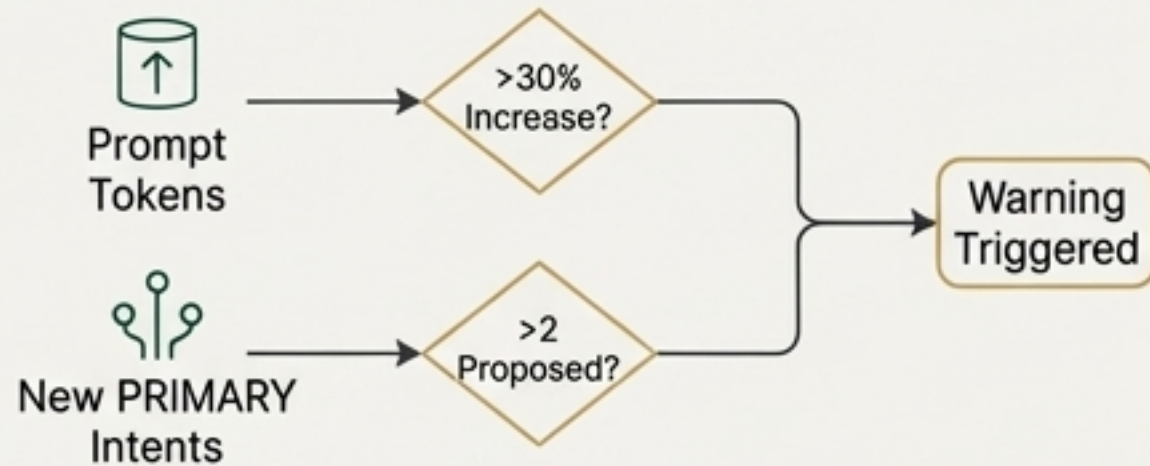
Safety and **stability** are not afterthoughts; they are core to the pipeline's design. The system **actively enforces rules to maintain a healthy intent structure**.

Guardrail 1: Fragmentation Prevention

Rule: Heavily penalizes the creation of new PRIMARY intents and excessive growth in prompt tokens.

How it Works:

- Calculates a `fragmentation_score = (primary_growth * 0.7) + (secondary_deviation * 0.3)`.
- Triggers warnings if prompt tokens increase by >30% or if more than 2 new PRIMARY intents are proposed.



****Result**:** 82-90% of all proposals across datasets were correctly identified as SECONDARY.

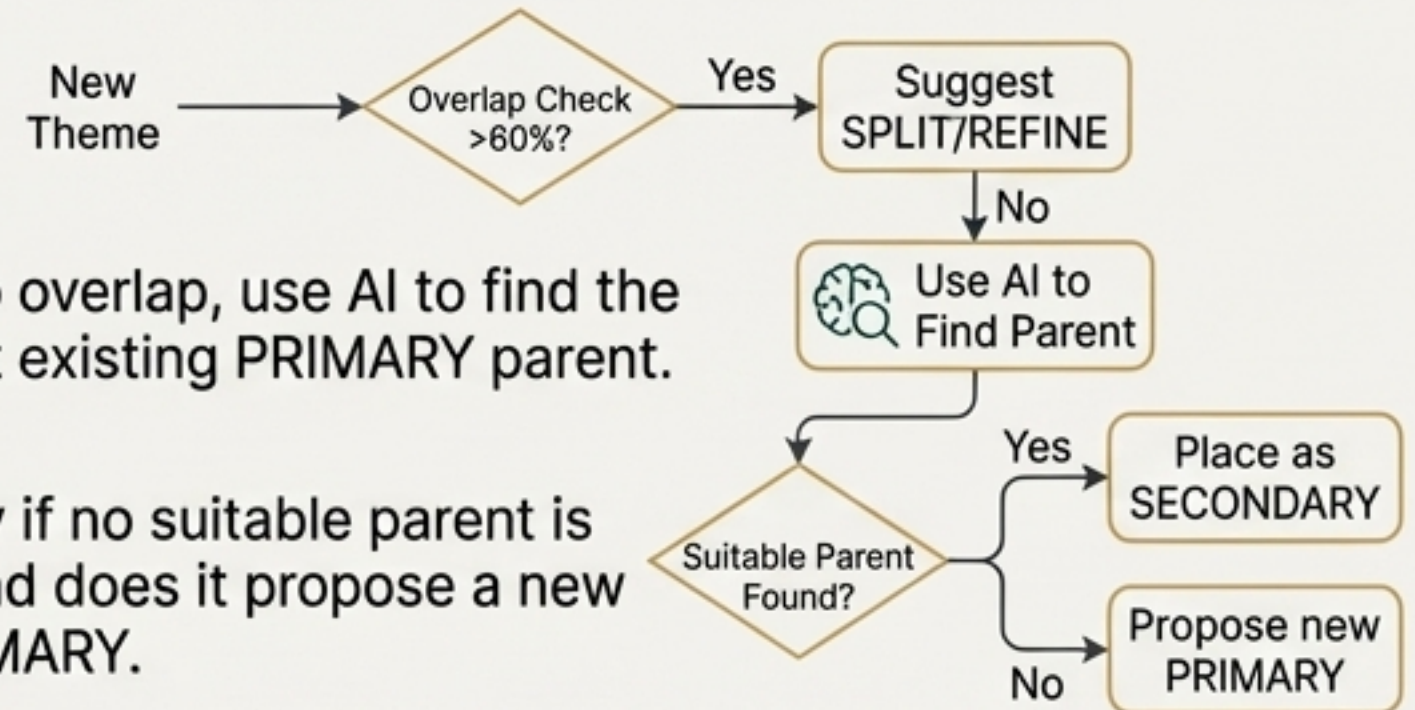


Guardrail 2: Hierarchical Placement Logic

Rule: Always prefer placing a new theme as a SECONDARY intent before considering it as a new PRIMARY.

How it Works:

- First, check for >60% overlap with existing intents to suggest a SPLIT or REFINE.



- If no overlap, use AI to find the best existing PRIMARY parent.

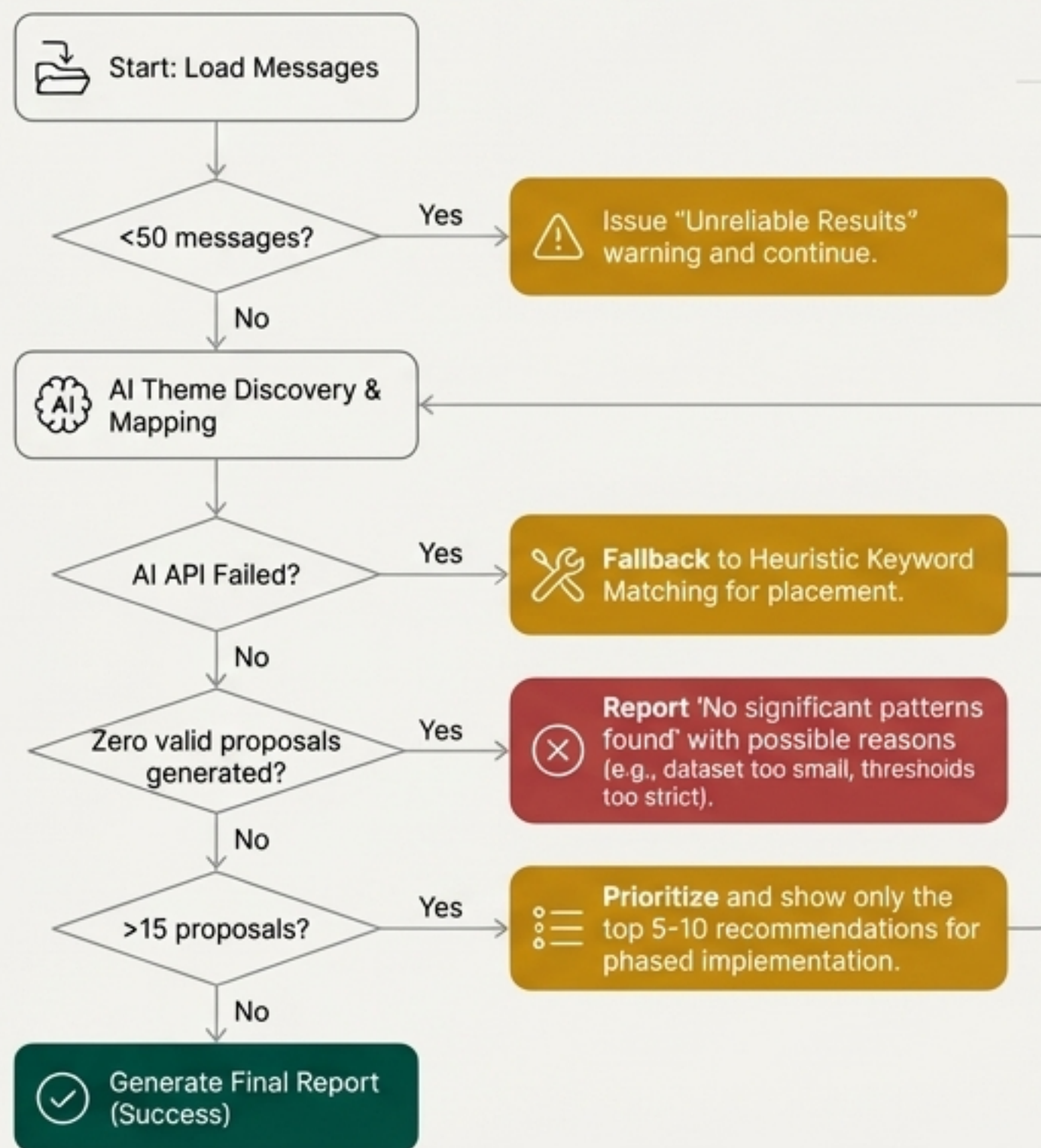
- Only if no suitable parent is found does it propose a new PRIMARY.

****Result**:** Ensures a logical, nested taxonomy and avoids a flat, confusing structure.



The System is Resilient by Design: Fallback Decision Logic

The pipeline follows a clear decision path to handle common issues, ensuring it always completes its run and provides a useful report.



V2 Pipeline: A Summary of Innovation and Proven Effectiveness

Core Innovations



AI-Driven Discovery: Scales across any domain without hardcoding.



Hierarchical Mapping: Prevents over-fragmentation by intelligently preferring SECONDARY intents.



Multi-Criteria Scoring: Provides transparent, evidence-based recommendations, not just suggestions.



Guardrails First Design: Actively prevents bad outcomes like prompt bloat and high fragmentation.



Actionable Output: Delivers clear guidance on what to implement, review, or ignore.

Effectiveness in Practice



Controlled Growth: 82-90% of all valid proposals were correctly placed as SECONDARY intents.



Actionable Insights: Identified 4 high-priority, implementable intents for each of the two test datasets.



Effective Guardrails: The token limit warning was correctly triggered on the Fragrance dataset (+39% > 30% threshold).



Acceptable Fragmentation: Both analyses resulted in acceptable fragmentation scores (0.21 and 0.34).



Resilient Operation: Fallback logic ensures the pipeline runs to completion even with API failures.